

Term Project: Milestone 5

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U.S. Childcare Costs: Economic Patterns and Affordability Challenges

Findings

The National Database of Childcare Prices dataset reveals important patterns in the childcare market in the United States. After resolving schema drift between years and re-indexing all dollar amounts to 2023 values, I built county-to-state aggregates and exported a tidy JSON to power an interactive HTML/JS dashboard.

In my EDA, I found a clear progression of childcare costs by age group, with infant care being the most expensive, followed by toddler care, preschool, and school-age care. Center-based care is consistently more expensive than family-based childcare across all age groups, with center-based infant care costing a median of \$134.50 per week nationally compared to \$106.00 for family-based care. The analysis confirms that median center-based infant care runs as high as \$303 per week in the District of Columbia, while preschool care generally falls to roughly two-thirds of infant care costs. The premium for center-based care over licensed family homes varies dramatically by geography—from just 10 percent in rural Arkansas to nearly 40 percent in suburban New Jersey.

There appears to be a strong positive correlation ($\rho \approx 0.62$) between median household income (MHI) and childcare costs. States with higher median household incomes like New Jersey (\$73,234), Connecticut (\$71,304), and Maryland (\$69,174) generally have higher childcare costs. This pattern is even more pronounced at the county level, where affluent counties in the DC metro area such as Loudoun County, VA (\$121,962) have significantly higher childcare costs.

Affordability, expressed as the share of median household income required for full-time infant care, averages 15.9 percent nationwide—well above the federal benchmark of 7 percent. Counties in the Boston-to-D.C. corridor routinely exceed 25 percent, whereas many Midwest counties stay under 10 percent. Notably, when price is divided by income, the relationship becomes almost flat or slightly regressive, indicating that middle-income families feel the greatest financial squeeze.

The data also reveals considerable geographic disparities. The Northeast and Mid-Atlantic regions generally have both higher incomes and higher childcare costs, while Southern states tend to have lower costs but also lower incomes. This suggests that while raw costs are lower in some regions, the proportion of income spent on childcare may still create significant financial strain for families.

Assumptions

My analysis assumes that the childcare options represented in the dataset are reasonably accessible to families in those areas, though actual availability isn't directly measured. Additionally, I assume that median household income is an appropriate metric to use when considering affordability, recognizing that many households have multiple income streams that contribute to childcare decisions.

Items That Still Need Clarification

Several aspects of the data warrant further investigation. First, the relationship between childcare quality and cost isn't captured in this dataset. Higher costs in certain areas might reflect higher quality care, regulatory differences, or simply market dynamics. Second, the dataset does not capture informal childcare arrangements, which might be significant in many communities.

Finally, the precise impact of local policy differences, such as subsidy programs or regulatory requirements, on childcare prices isn't clear from the data.

Direction of Story

The story focuses on how geographic and economic disparities create vastly different childcare realities for families across the country. I highlight the percentage of household income required for childcare in different regions and demonstrate how this creates a disproportionate burden on middle and lower-income families. The core message is that closing geographical affordability gaps is as critical as lowering the national average price, as childcare costs represent a critical economic issue that affects workforce participation, family financial stability, and early childhood development opportunities.

Target Audience

The primary audience for this analysis will be policymakers and community leaders at the state and local levels who have influence over childcare subsidies, regulations, and economic development policies. This audience is likely familiar with economic data but may benefit from clear visualizations showing how childcare costs impact their specific constituencies. They'll be particularly interested in comparative data showing how their region compares to others and what policy levers might improve affordability.

Mediums to Include

1. **Interactive Dashboard:** A geographic-based interactive visualization showing childcare costs as a percentage of median household income by county, with filtering capabilities for different age groups and care types. This enables policymakers to zoom from a

national choropleth to an individual county, slice by age group or provider type, and instantly see affordability percentages and historical trends.

2. **Infographic:** A visually striking narrative that distills key facts into an executive-ready format for social media or agency newsletters. It provides a "ladder" graphic that compares weekly childcare costs, highlighting the affordability gap across different economic brackets and emphasizing the impact on workforce participation and financial stability for families.
3. **Presentation Deck:** A seven-slide PowerPoint deck with key metrics and charts showing the progression of childcare costs across geographic regions, designed for in-person briefings with stakeholders and community forums. Helps non-technical audiences absorb the progression from data to implications to policy options.

Design Decisions

Blue-to-magenta gradients were chosen because they are most accessible, even accommodating color-blind palettes, and help to emphasize the outliers at both low and high ends. The midpoint is centered on the 7 percent federal affordability target to cue viewers instantly to counties in "crisis red." My charts adopt a left-aligned title hierarchy and ample white space to reduce cognitive load.

Ethical Considerations

Only minimal transformations were applied to the data: missing county-year observations (<3 percent) were forward-filled from the nearest prior year to avoid visually misleading blanks, and all monetary values were converted to 2023 dollars using CPI-U. These changes are flagged in the methods section of the dashboard and footer of every graphic to maintain transparency.

The dataset contains no personally identifiable information, so FERPA or HIPAA do not apply; however, presenting county averages risks masking intra-county inequities. To mitigate that, narrative copy repeatedly reminds readers that affordability challenges can be sharper for single-parent and low-income households than aggregate figures suggest. A legend note discloses that informal care is excluded. All data was obtained from a publicly funded federal source (U.S. Department of Labor's National Database of Childcare Prices), ensuring the ethical sourcing and verification of data credibility.

Lessons Learned

Next time I would prototype the HTML/JS dashboard earlier. I would also budget more time to merge subsidy-rate tables so that cost and public-funding gaps can be shown side-by-side. On the positive side, I really enjoyed translating dense data into user-friendly narratives that stakeholders can immediately recognize and understand. This experience reinforced the importance of not just presenting data accurately but presenting it in a way that resonates with the audience's real-world concerns and experiences.

References

1. U.S. Department of Labor, Women's Bureau. (2024). *National Database of Childcare Prices* [Data set].
<https://content.bellevue.edu/cst/dsc/640/2024Update/Datasets/nationaldatabaseofchildcareprices.xlsx>
2. U.S. Department of Labor, Women's Bureau. (2024). *National Database of Childcare Prices*. <https://www.dol.gov/agencies/wb/topics/featured-childcare>
3. U.S. Department of Labor, Women's Bureau. (2020, October). *National Database of Childcare Prices: Final report* (Technical guide).
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